

A Comparative Analysis of Crime and Unemployment Pre and Post-Pandemic in the United States

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Abstract—The emergence and widespread effects of COVID-19 has had, and continues to have, a profound impact on both the healthcare outcomes of humans and worldwide economies. Governments throughout the world have responded with stimulus payments to safeguard their citizens’ safety and basic needs along with injections to their economies in order to prevent mass economic decline.

Nevertheless, future economic outcomes remain uncertain. Due to inflation, supply-chain delays, and various other ailments (e.g. the H5N1 avian flu); the price of basic goods has increased while wages have remained stagnant. All market indications point to a global recession. Yet as of January 2023, the unemployment rate is at 3.4%, which is the lowest since May 1969.

To gain further insights into current market outcomes, our study aims to collect data from pre and post COVID-19 (2018-2023). The dataset will focus primarily on the unemployment rate in the United States. This metric will be compared against the stock valuation of the Dow Jones Industrial Average (DJIA), NASDAQ Composite, and S&P 500 (Standards and Poor’s). Additionally; we will also look at other economic indicators such as the frequency of job postings on LinkedIn, and documented criminal offenses during this period. This will allow us to conclusively determine the multifaceted impact of the COVID-19 pandemic on the country’s economy.

Index Terms—COVID-19, pandemic, crime, unemployment, stock market, job hiring

I. INTRODUCTION

The COVID-19 pandemic has left an enduring mark on the global landscape, disrupting the daily lives of millions and challenging the very foundations of public health and economic systems. As the pandemic unfolded, countries worldwide implemented various measures to curb the spread of the virus and protect their citizens. In the United States, and many other nations, measures such as social distancing, travel and work restrictions, and even varying periods of lockdowns, led to significant economic downturn and unemployment that peaked above the levels of the 2008 Recession.

In attempts to curb the crisis, the Trump, later Biden admin, along with local State governments implemented series of stimulus packages & policies aimed at invigorating the economy and providing financial support to individuals and businesses. These programs, such as the CARES Act

passed at the height of the pandemic under the Trump admin, which included the Payment Protection Program (PPP), stimulus checks, unemployment compensation, and, mortgage forbearance, gave much-needed relief. Nevertheless, despite these efforts and recent improvements in unemployment rates, reaching a low of 3.4% in January 2023, a myriad of economic issues persist, creating an uncertain landscape for many Americans.

In our current post-pandemic era, inflation has emerged as a significant concern. This is driven by the pandemic’s supply chain disruptions, increased demand for certain goods and services, as well as the consequences of the government’s well-intentioned but shortsighted fiscal policies. The lax verification for PPP loans and unemployment insurance, which were aimed at covering payroll or providing funds for those unable to find employment, allowed for fraud and abuse to occur [1][2] with billions being stolen. Recent reports also point to instances of corporate greed, with some companies intentionally and unnecessarily raising prices on goods [3]. These factors have contributed to the rising cost of living, with the prices of basic and essential goods increasing at an alarming rate. Simultaneously, wages have largely remained stagnant, exacerbating the financial struggles faced by many households.

Furthermore, global events such as the emergence of the H5N1 avian flu and the ongoing war of Russian aggression in Ukraine have further added to the economic uncertainty, with potential implications for food insecurity and crime[4], international trade, and public health. The pandemic has also exposed vulnerabilities in the global supply chain, leading to delays and shortages in various industries, which in turn have affected consumer behavior and corporate performance.

As the United States wades through these complex challenges, it is essential to understand the multifaceted impact this pandemic has and continues to inflict. Particularly the relationship between crime and unemployment. Our study also seeks to provide a comparative analysis of pre and post-COVID-19 and compare the circumstances against the 2008 Recession. We will also assess the relationship between unemployment and stock market performance focusing on the Dow Jones Industrial Average (DJIA), NASDAQ Composite, and S&P

500 (Standards and Poor's) indices. We will also examine job hiring data provided by LinkedIn to gauge and provide possible explanations for which industries were impacted the least and most, respectively. Our study aims to provide valuable insights into the consequences of the pandemic and its implications for the future of the American economy.

II. METHOD

Python and R languages, with accompanying packages such as Pandas and NumPy were used along with Plotly and Matplotlib for the creation of graphs. MySQL database was used to contain the data when appropriate or was read directly from the .CSV file.

The various different datasets used will be expanded on in the sections below. a brief overview of the datasets used:

- California Unemployment Data
- California Crime Data
- LinkedIn Hiring Data
- Stock Market Historical Data
 - 1) DJIA
 - 2) NASDAQ Composite
 - 3) S&P 500

A. California Unemployment

We were able to examine the unemployment rates in California using a data explorer created by the State of California Employment Development Department[5] that allowed us to export the data as a CSV file. The CSV file included labor data for every county in California on the first day of the first month every year from 1976 until present time. First, we imported the data into RStudio and used the R programming language to perform exploratory data analysis. We utilized R's tidyverse library, which is specifically designed for data manipulation and analysis. The data had 11 columns and 23,700 entries. The first set of columns included area type (which all had the value 'County'), area name (name of county), date (in the 'mm/dd/yyyy' format), year, month (which all had the value 'January'), seasonally adjusted (either Y for yes and N for no), status (either preliminary or final), labor force, employment, unemployment, and unemployment rate. We extracted the area name, year, month, and unemployment rate columns, and did not take the 'seasonally adjusted' and 'status' columns into account. We decided to choose only entries where the 'seasonally adjusted' value was 'N'. This process allowed us to focus on the relevant data for our analysis.

B. California Crime

In this study, we examined the crime rates in California using the state's Department of Justice's database[6]. Initially, we attempted to use the FBI's Crime Data Explorer to obtain crime data for all states in the US. Unfortunately, we found that the data for certain years in some states were missing, which would have compromised the study's integrity. As a result, we decided to focus on California, where we found a comprehensive crime database provided by the state's own Department of Justice. To obtain the data, we downloaded the

	Area Type	Area Name	Date	Year	Month	Seasonally Adjusted (Y/N)	Status (Preliminary / Final)	Labor Force	Employment	Unemployment	Unemployment Rate
1	County	Alameda County	12/01/2022	2022	December	N	Preliminary	831100	808400	22700	0.027
2	County	Alpine County	12/01/2022	2022	December	N	Preliminary	580	550	20	0.042
3	County	Amador County	12/01/2022	2022	December	N	Preliminary	14170	13580	580	0.041
4	County	Butte County	12/01/2022	2022	December	N	Preliminary	92600	88900	3700	0.040
5	County	Calaveras County	12/01/2022	2022	December	N	Preliminary	21800	21210	590	0.032
6	County	Colusa County	12/01/2022	2022	December	N	Preliminary	10720	9410	1310	0.122
7	County	Contra Costa County	12/01/2022	2022	December	N	Preliminary	554700	538400	16200	0.029
8	County	Del Norte County	12/01/2022	2022	December	N	Preliminary	9240	8810	440	0.047
9	County	El Dorado County	12/01/2022	2022	December	N	Preliminary	92500	89900	2600	0.028
10	County	Fresno County	12/01/2022	2022	December	N	Preliminary	450400	422300	28100	0.062
11	County	Glenn County	12/01/2022	2022	December	N	Preliminary	12710	12060	650	0.052
12	County	Humboldt County	12/01/2022	2022	December	N	Preliminary	60500	56500	2000	0.034
13	County	Imperial County	12/01/2022	2022	December	N	Preliminary	70900	60400	10500	0.148
14	County	Inyo County	12/01/2022	2022	December	N	Preliminary	8240	7980	260	0.031
15	County	Kern County	12/01/2022	2022	December	N	Preliminary	388900	363000	25900	0.067
16	County	Kings County	12/01/2022	2022	December	N	Preliminary	56500	52700	3800	0.068
17	County	Lake County	12/01/2022	2022	December	N	Preliminary	28130	26700	1430	0.051
18	County	Lassen County	12/01/2022	2022	December	N	Preliminary	8890	8530	360	0.041
19	County	Los Angeles County	12/01/2022	2022	December	N	Preliminary	4927700	4712000	215500	0.044
20	County	Los Angeles County	12/01/2022	2022	December	Y	Preliminary	4933800	4701500	232300	0.047
21	County	Madera County	12/01/2022	2022	December	N	Preliminary	63600	59800	3800	0.060
22	County	Marin County	12/01/2022	2022	December	N	Preliminary	126800	120000	6800	0.052
23	County	Mariposa County	12/01/2022	2022	December	N	Preliminary	6660	6360	300	0.045

Fig. 1. Table of Labor Data from California Employment Development Department.

Fig. 2. California Employment Development Department Data Explorer.

Criminal Justice Data in the Crimes and Clearances (including Arson) section. The section included both a monthly CSV file and an informational document with details on the data. We conducted several processes to analyze the crime data in California. After importing the data into RStudio, we used the tidyverse library to do our analysis. The crime data we analyzed contained 71 columns and 324,615 entries for every county in California from January 1985 to October 2021 at the time of this research study. The first four columns of the data were year, county, NCICCode, and month. The second set of columns included the total number of violent and property crimes reported, as well as the total number of specific crimes in these categories. The third set of columns included the total number of clearances reported for the previously mentioned crimes. The fourth set included columns for the total number of different arsons committed, as well as the grand total of all arsons reported and cleared. The remaining set included columns for supplemental crime information reported. To conduct our analysis, we extracted specific columns relevant to our study, which were the different types of crimes reported. We also selected rows where the month had a value of 1, as we needed to join this data with our California unemployment data, which only contained data for the first day of the first month of every year. We then cleaned the data by replacing any missing values with 0. We then joined the crime data

with the California unemployment table based on year, county, and month. After joining the data, we converted the year and month to a specific datetime format (yyyy-mm-dd), which is the required format for visualizing changes in crime rates over time. We also highlighted three significant periods in the chart we created, corresponding to the 2008 Recession (2007-12-01 - 2009-06-01), Pre-COVID (2017-01-01 - 2020-01-01), and COVID-19 Pandemic (2020-01-01 - 2021-12-01). These periods were highlighted to compare crime rates between them, with the goal of determining if there were any significant differences in crime rates during these periods. Additionally, we plotted scatterplots between unemployment and the total number of violent crimes reported, as well as the total number of property crimes reported. We then calculated the Pearson’s product-moment correlation between each crime type and unemployment rate for all three periods. The correlation was calculated using R’s built-in correlation test function, which used 17 degrees of freedom. This analysis was conducted to determine if there were any significant correlations between unemployment rates and crime rates during these periods.

The screenshot shows the 'OPENJUSTICE' data portal. It has tabs for 'DATA EXPLORATION', 'DATA PORTAL', and 'RESOURCES'. Under 'DATA EXPLORATION', there are sections for 'Crimes and Clearances (including Arson)', 'Death in Custody & Arson-Related Deaths', 'Domestic Violence Related Calls for Assistance', and 'Sex Violence Reporting Orders'. Each section contains a brief description and a 'Summary' link. For example, 'Crimes and Clearances' has a summary of 124,144, 'Death in Custody' has 1,141, 'Domestic Violence' has 16,141, and 'Sex Violence' has 10,141.

Fig. 3. California Department of Justice Data Portal.

The table displays crime and clearance data for various counties in California. The columns are: Year, County, NCCCode, Month, Violent sum, Homicide sum, Forfeiture sum, Robbery sum, AppAssault sum, Property sum, Burglary sum, and VehicleTheft sum. The data is organized by year and county, with rows showing monthly totals. For example, in 1995, Alameda County had 1 violent sum, 0 homicide sum, 0 forfeiture sum, 0 robbery sum, 0 appassault sum, 0 property sum, 0 burglary sum, and 0 vehicletheft sum.

Fig. 4. Table of Crimes and Clearances from California Department of Justice.

C. LinkedIn Job Hiring

A web scraper was utilized to gather the data directly from LinkedIn’s website[7]. The data was separated by months, each on its own page. We manually located all links associated with the United States and the appropriate time frame (2019 - 2023) and compiled them into an array. These were then fed into our web scrapper utility using Pandas to extract the

needed table data from each page and then saved as .csv. At this stage, each .csv contained its own month (Figure 5).

Table 1: Hiring on LinkedIn, by Industry, through January 2023

Industry	Jan-22	...	Oct-22	Nov-22	Dec-22	Jan-23	MoM% Change	YoY% Change
Accommodation	1.15	...	1.01	0.99	1.00	0.95	-5.2	-17.3
Administrative and Support Services	1.34	...	1.08	1.07	1.04	1.03	-0.5	-23.3
Construction	1.44	...	1.20	1.14	1.14	1.18	+3.1	-18.4
Consumer Services	1.26	...	1.13	1.10	1.09	1.07	-1.7	-15
Education	1.28	...	1.22	1.19	1.17	1.13	-3.5	-11.7
Entertainment Providers	1.20	...	0.97	0.91	0.92	0.92	0	-23.5

Fig. 5. Example table from LinkedIn.

Next, we created used data wrangling techniques. Only the last month, which contained the most up-to-date information – a month-over-month change in percentage points, of each table was used, all others were discarded. Any discrepancies, such as unneeded data or categories that didn’t span our full timescale were discarded (Figure 6).

Agriculture	0.77	1.45	0.93	1.11
Arts	0.48	1.06	0.61	0.71
Consumer Goods	0.6	1.12	0.81	0.94
Corporate Services	0.67	1.53	0.88	0.95
Design	0.52	1.16	0.75	0.84
Energy & Mining	0.74	1.33	0.95	1.07
Entertainment	0.49	1.19	0.7	0.77
Finance	0.64	1.19	0.8	0.95

Fig. 6. Final CSV file.

All categories were then initialized into a dictionary and then all .CSV files were examined to gather the relevant data and combined to output one .CSV file and feed into the MySQL database.

A formula to sum for relative percentage increase was utilized to discover the least and most job hirings. This Allowed categories to be sorted from least to greatest in terms of hires.

$$GrowthIndex_{xt} = \sqrt[12]{\prod_{i=1}^{39} TupleA_{xt,i}}$$

D. Stock Market Data

Data for the stock market indices were gathered and compiled in a similar fashion to LinkedIn’s. Instead of web scraping, we went to the page for each index and loaded in the browser to the past 20 years, and then printed the webpage as a PDF file. Then, using Pandas, the PDF files were scrapped for data (Figure 7).

Only the “month” and “close” values were kept from the table, all other values were removed. Numbers in scientific notation were found using regex and converted to their full value

DJI

Dow Jones Industrial Average Index
Price History & Data

\$31,834.24 -412.31 (-1.28%) 3:38 PM 03/17/23

Dow Jones Indices | USD | Realtime

Charting

Historical Prices



Historical Prices

Month Starting	Open	High	Low	Close	Change %	Avg. Volume
Mar. 01, 2023	32.66K	33.57K	31.43K	32.25K	-1.26%	350,920,853
Feb. 01, 2023	34.04K	34.33K	32.64K	32.66K	-4.19%	323,028,798
Jan. 03, 2023	33.15K	34.34K	32.81K	34.09K	2.83%	341,196,523
Dec. 01, 2022	34.53K	34.71K	32.57K	33.15K	-4.17%	332,597,041
Nov. 01, 2022	32.86K	34.59K	31.73K	34.59K	5.67%	347,847,739
Oct. 03, 2022	28.86K	32.89K	28.70K	32.73K	13.95%	372,536,262
Sep. 01, 2022	31.45K	32.49K	28.73K	28.73K	-8.84%	373,402,150
Aug. 01, 2022	32.76K	34.28K	31.51K	31.51K	-4.06%	305,453,303
Jul. 01, 2022	30.72K	32.91K	30.14K	32.85K	6.73%	322,198,322
Jun. 01, 2022	33.16K	33.27K	29.66K	30.78K	-6.71%	390,273,730

Fig. 7. DJIA Stock Market Data from SeekingAlpha.com saved as PDF.

(Figure 8). These files were then fed into the MySQL database and retrieved with Python. All datasets were normalized and then graphed to be compared against the unemployment rate. Time series relating to the 2008 Recession (Jan 2007 - Dec 2009) and peri-pandemic (Jan 2020 - Dec 2021) were sliced to have their values assessed using Pearson's correlation coefficient (PCA) function available from NumPy.

Month Starting	Close
Mar. 01, 2023	32250
Feb. 01, 2023	32660
Jan. 03, 2023	34090
Dec. 01, 2022	33150
Nov. 01, 2022	34590
Oct. 03, 2022	32730
Sep. 01, 2022	28730
Aug. 01, 2022	31510
Jul. 01, 2022	32850

Fig. 8. DJIA Stock Market Data as .CSV.

III. RESULTS

A. Unemployment and Crime in California

Our study found an inverse relationship between crime and the unemployment rate between both periods, the 2008 Recession and the COVID-19 pandemic. That is, as the unemployment rate rose, the total amount of reported crime fell (Figure 11).

However, one closer look at all crime categories, certain types of crimes rose in this pandemic period while others fell. In our dataset for violent crime: homicide and aggravated assault rose, while robbery fell. This mostly coincides with our 2008 Recession data. It is important to note that reporting of rape increased post-2015 – as such, this data point will be excluded from our results due to the mismatch of data between the two periods. Social and legal events (such as #MeToo) (Figure 12).

Follow
23.17K followers

As for property crimes, again, certain categories of crimes rose: vehicle theft and arson, while larceny and burglary fell. This deviates from the 2008 Recession dataset which shows and increase in burglaries while the other categories are trending downwards. (Figure 13).

The Results of our PCA do corroborate the visual findings as seen on the graph (Figure 9, 10).

Crime	Pearson's r	p-value
Robbery	-0.5318351	0.007477
Larceny	-0.636386	0.0008284
Arson	0.5530151	0.005066

Pearson's correlation coefficients and p-values at $p < 0.05$ for Total Reported Crimes and Unemployment Rate during the COVID-19 Pandemic

Fig. 9. Pearson's Correlation Coefficient during COVID19 Pandemic.

Period	Crime	Pearson's r	p-value
The Great Recession (Dec 2007 - Jun 2009)	Homicide	-0.51	0.03
	Vehicle Theft	-0.84	7.505e-06
	Larceny Theft	-0.63	0.004
Pre-COVID (Jan 2017 - Dec 2019)	Burglary	0.67	4.677e-06
	Vehicle Theft	0.73	2.294e-07
The COVID-19 Pandemic (Jan 2020 - Dec 2021)	Robbery	-0.53	0.007
	Larceny	-0.64	0.0008
	Arson	0.55	0.005

Pearson's correlation coefficients and p-values at $p < 0.05$ for Total Reported Crimes and Unemployment Rate

Fig. 10. Pearson's Correlation Coefficient by period.

B. LinkedIn Hiring Data

The results from the LinkedIn hiring data spread over 18 in one graph follow the trends of market instability during the start of the pandemic which then tapers out.

Nevertheless, it does highlight the obvious industries that were impacted by the switch to remote work. Office jobs or those works that can be seldom performed were able to continue hiring, while more traditional jobs requiring social or physical interactions in confined spaces saw less job hiring.

C. Stock Market

The effects of the 2008 Recession and the COVID-19 Pandemic are in stark contrast to each other. 2008 Saw a collapse of our indices and an exponential increase in unemployment but during the pandemic, unemployment soared far above the 2008 Recession. Yet, our stocks saw minor dips from which

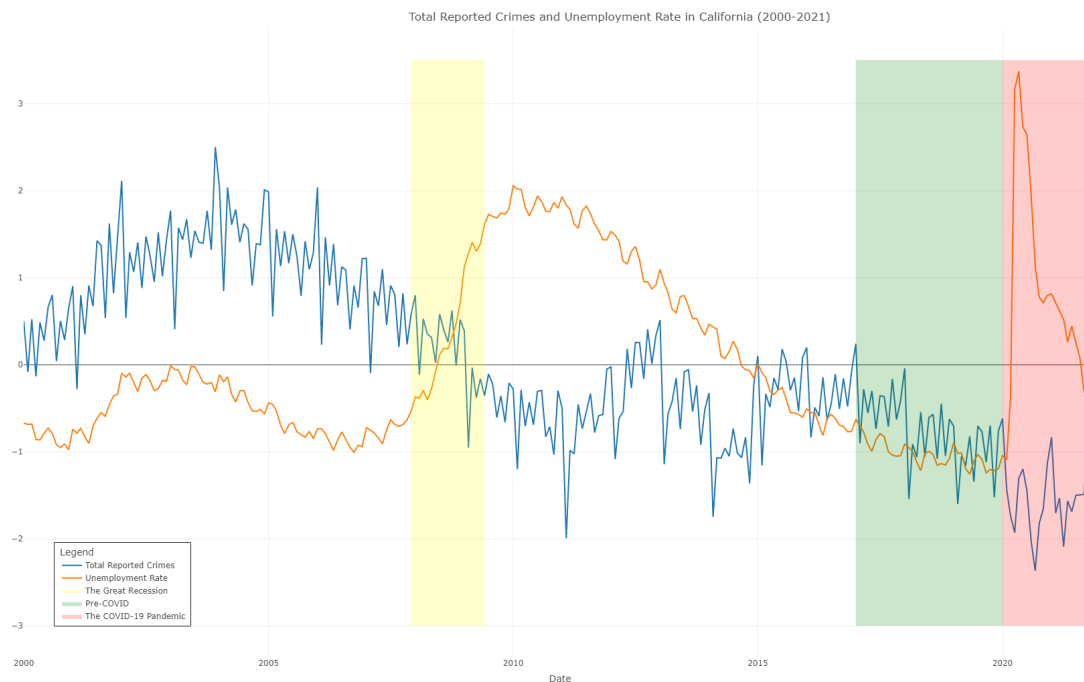


Fig. 11. Total Reported Crimes and Unemployment Rate in California (2000 - 2021).

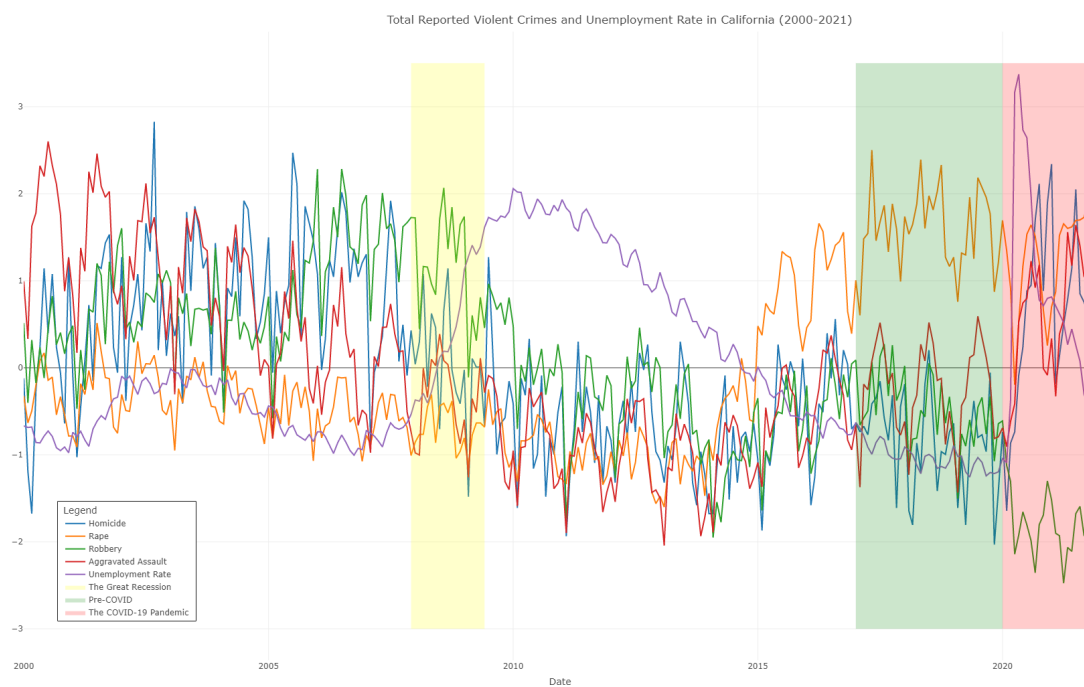


Fig. 12. Total Reported Violent Crimes and Unemployment Rate in California (2000 - 2021).

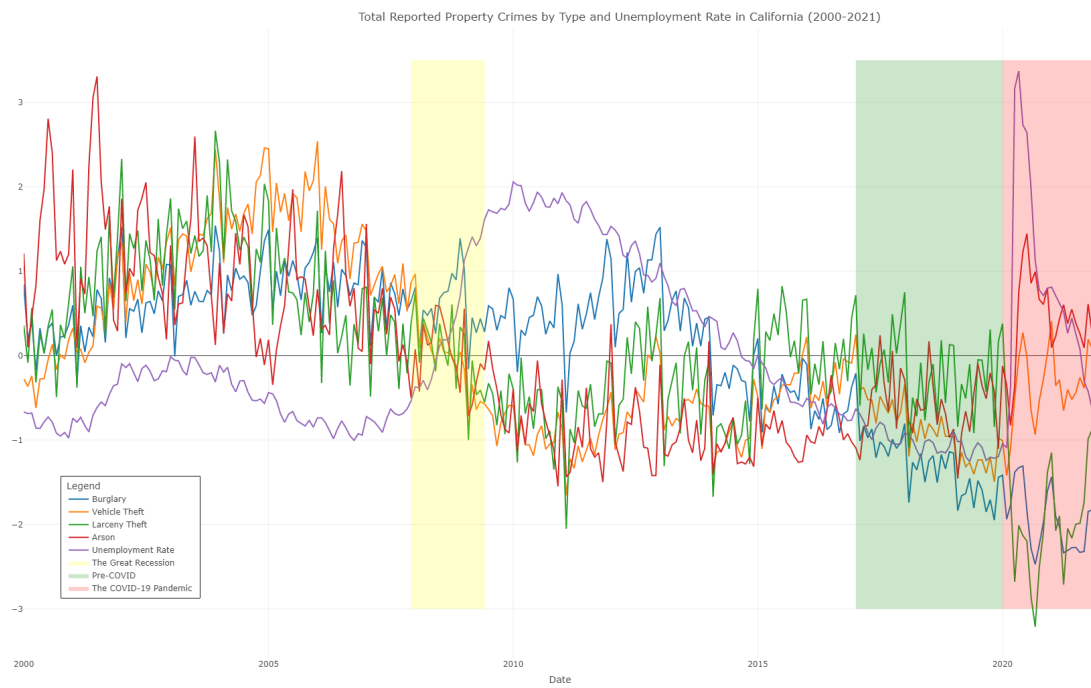


Fig. 13. Total Reported Property Crimes and Unemployment Rate in California (2000 - 2021).

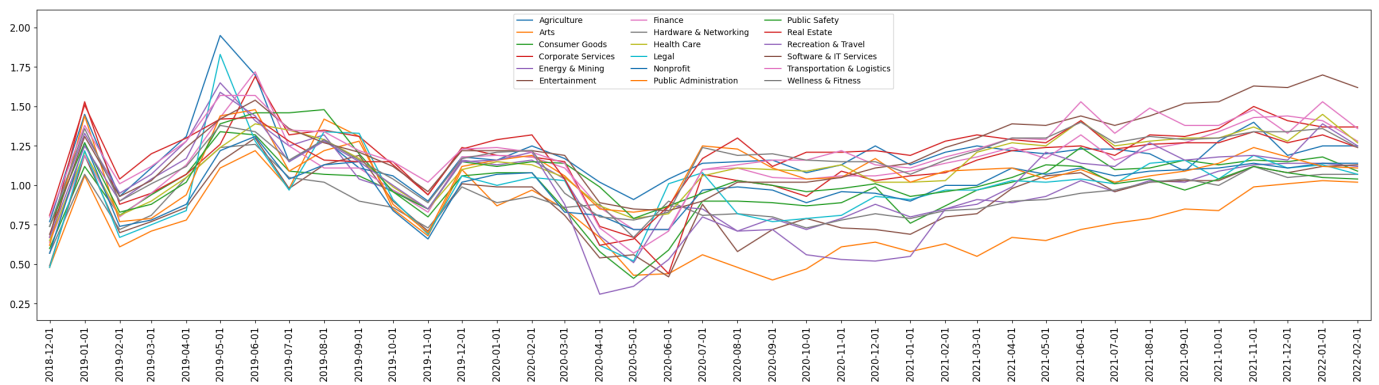


Fig. 14. All Categories of LinkedIn Hiring During the COVID-19 Pandemic (Dec 2018 - Feb 2022).



Fig. 15. Linked Hiring: Highest Job Hiring During the COVID-19 Pandemic (Dec 2018 - Feb 2022).

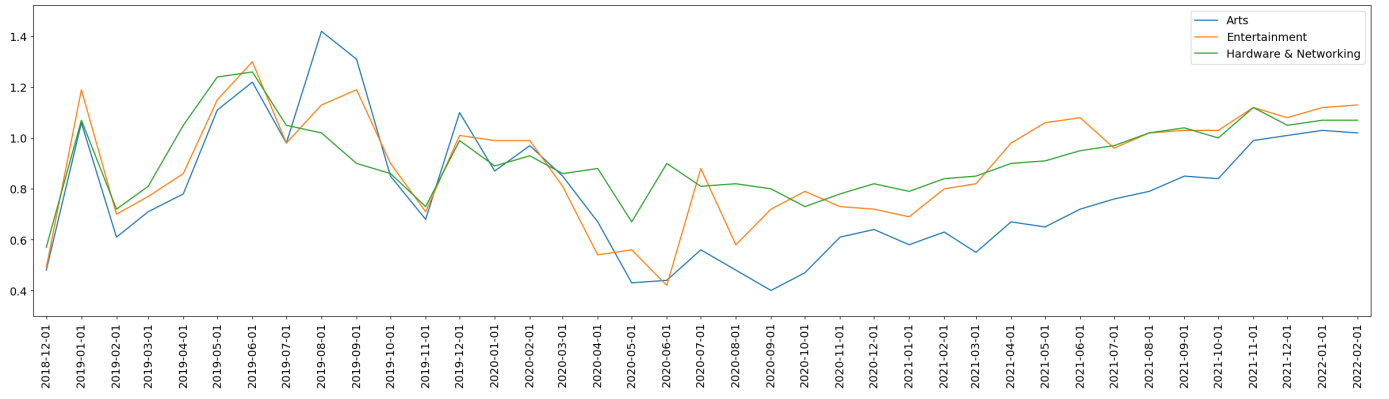


Fig. 16. Linked Hiring: Lowest Job Hiring During the COVID-19 Pandemic (Dec 2018 - Feb 2022).

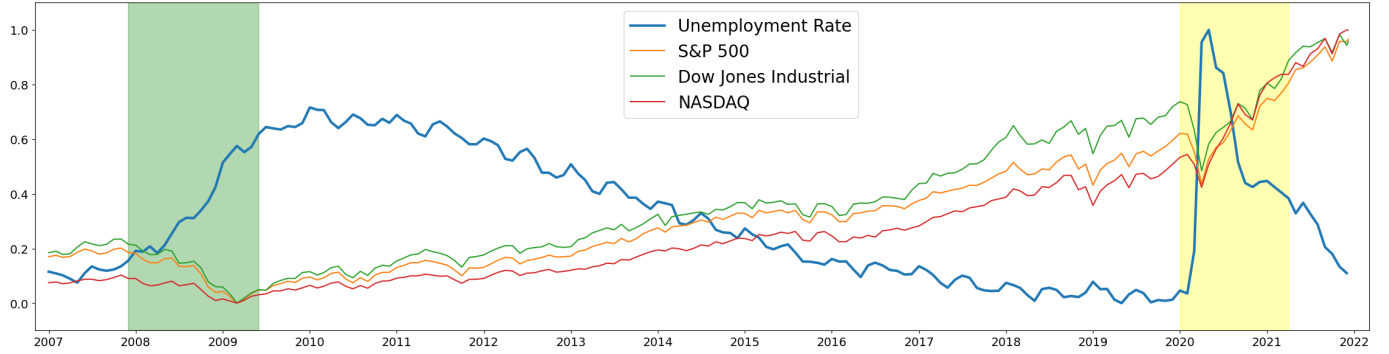


Fig. 17. Stock Market Index against Unemployment (2007 - 2022).

they quickly bound back, achieving all-time highs while in the midst of the pandemic with still high unemployment rates.

The PCA analysis of these two periods shows the effects of each circumstance.

TABLE I
STOCK PERFORMANCE DURING RECESSION & COVID-19

Stock Index	2008 Recession	COVID-19 Pandemic
S&P 500	-0.8986	-0.6205
DJIA	-0.8949	-0.6812
NASDAQ	-0.7839	-0.5109

IV. CONCLUSION

Based on the findings of our study, we can conclude that the relationship between unemployment and crime in California is complex and nuanced. While our study found an overall inverse relationship between crime and the unemployment rate, certain types of crimes, such as homicide and aggravated assault, showed an increase during the COVID-19 pandemic. Similarly, the impact of market instability on job hiring was felt more strongly in industries that require social or physical interactions in confined spaces, while office jobs were able to continue hiring.

Additionally, the stock market showed a different pattern of response to the 2008 recession and the COVID-19 pandemic,

with the latter experiencing a sharp increase in unemployment but a quick rebound in the stock market. Our findings suggest that the impact of economic crises on different aspects of society can vary and depend on various factors.

It is important to note that our study has some limitations, such as relying on reported crime data, which may not accurately reflect the actual crime rate, and not accounting for other potential factors that could influence the relationship between unemployment and crime. Future studies could address these limitations by using more comprehensive data and exploring other factors.

Overall, our study provides insights into the complex relationship between economic crises, unemployment, and various societal factors, highlighting the need for continued research and exploration in this area.

There is a lot of potential for further research on the relationship between unemployment and crime, particularly in light of the COVID-19 pandemic and its unique economic impact. Future work could explore more deeply the reasons behind the inverse relationship between the unemployment rate and reported crime, and why certain types of crime may increase or decrease during times of high unemployment. In addition, expanding the dataset to include more recent data could help shed light on any longer-term trends. As for the LinkedIn hiring data, it could be interesting to look at how hiring patterns have shifted as the pandemic has progressed and

remote work has become more prevalent. Finally, while our study found that the stock market rebounded quickly despite high unemployment rates during the COVID-19 pandemic, further analysis could be done to explore this relationship in more detail and understand why the stock market reacted differently during the 2008 Recession.

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